MIL-I-7063B
20 April 1979
SUPERSEDING
MIL-I-7063A
21 November 1974

MILITARY SPECIFICATION

INDICATOR, POSITION, RUDDER TRIM TAB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of rudder trim tab position indicator.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-T-5583	Transmitter, Position, 28 Volts DC
MIL-I-7062	Indicator, Position, Control Surfaces, 28 Volt DC, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Markings of U S Military Property
MIL-STD-794	Parts and Equipment, Procedures For Packaging and Packing of
MS28002	Indicator - Position, Control Surfaces, 1-3/8 Inch Dial, 28 Volts DC
MS33680	Insert Arrangements, MIL-C-5015 Electric Connector, Sizes 8 and 10

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The requirements specified in Specification MIL-I-7062 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.1.1 Conflicting requirements. In the event the requirements of this specification and the general specification conflict, the requirements of this specification shall govern.

3.2 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.3 Materials. Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein, shall be of the best quality and entirely suitable for the purpose intended.

3.4 Design and construction. The indicator shall be designed for operation with a position transmitter in accordance with Specification MIL-T-5583.

3.4.1 Resistance. The direct-current resistances of the indicator measured between the connector pins A, B, and C shall be 475± 100 ohms. Deviation from the average of the resistance values between the three coils shall not exceed ±1-1/2 percent.

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3.4.2 Pointer. The pointer shall conform to Specification MIL-I-7062. The length of the pointer shall be such that the tip of the pointer shall extend into the scale from 1/3 to 2/3 the length of the shortest graduation.

3.4.3 Electrical connector. The indicator shall be provided with a 10 SI-3P electrical connector as specified on Standard MS33680 and in accordance with Specification MIL-I-7062 for hermetically sealed indicators.

3.4.4 Indicator case. The case shall conform to Standard MS28002. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the indicator case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.4.4.1 Filling medium. The case shall be filled with helium of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dewpoint -65° C) at the filling pressure. The absolute pressure of the helium in the case shall be approximately 1 atmosphere.

3.4.5 Dial. The dial shall conform to figure 1. Graduations shall be equivalent to 5 indicated degree intervals. The dial scale plate shall be securely fastened to the frame of the mechanism. The minimum diameter of the dial shall be 1-5/16 inches when measured across the outside ends of the graduations.

3.4.6 Markings. The dial markings shall be as shown on figure 1, and shall be durable to withstand usage encountered in service.

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.5.1 Use of AN or MIL designations. AN or MIL designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence or sales matter,

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until notification has been received from the qualifying activity that the product has been approved for aeronautical use.

3.6 Workmanship.

3.6.1 General. The indicator, including all parts and accessories, shall be finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, painting, alinement of parts and tightness of assembly screws and bolts; etc.

3.6.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.6.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage to the screw or bolt or threads.

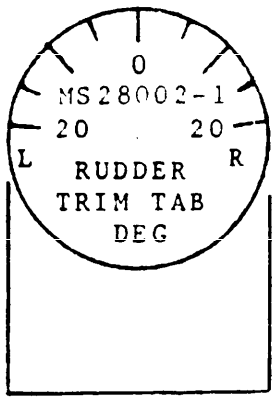
3.6.4 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling, inspection, and test procedures. Methods of sampling, inspection, and tests shall be as specified in Specification MIL-I-7062.

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1-3/8 APERTURE

MARKING	HEIGHT OR LENGTH ± 0.010	WIDTH OF LINE OR GRADUATION ± 0.005	MATERIAL OR FINISH
LETTERING "RUDDER"	0.094	0.020	
LETTERING "L" "O" "R"	0.125	0.020	
LETTERING "TRIM TAB DEG"	0.062	- - -	FLUORESCENT
NUMERAL "20"	0.094	0.020	LUMINESCENT
MAJOR GRADUATIONS	0.156	0.031	
MINOR GRADUATIONS	0.125	0.020	
SHADED PORTION OF POINTER	- - -	- - -	
LETTERING PER MS28002-1	0.062	0.010	DURABLE
BACKGROUND OF DIAL	- - -	- - -	DULL BLACK

DIMENSIONS IN INCHES

FIGURE 1 DIAL AND MARKINGS

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* 4.3 Packaging inspection.

* 4.3.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

* 4.3.2 First article. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

* 5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified.

* 5.1.1 Level A.

* 5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

* 5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

* 5.1.1.3 Preservation application. Not required.

* 5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method 1A-8 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.3 of this specification.

* 5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

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* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601 plywood box, shall be used for the shipping container.

* 5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather-resistant.

* 5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

* 5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The rudder trim tab position indicator covered by this specification is intended for use with a position transmitter, conforming to specification MIL-T-5583, in indicating the position of the aircraft component in degree travel from the neutral position, and the resulting reaction on the aircraft attitude such as tending to turn the nose right or left.

6.1.1 For the trim tabs, the AN5786 position transmitter is connected to the tab actuating mechanism so that a 2 to 1 movement ratio is provided. The transmitter shaft shall be rotated 2 mechanical degrees for each degree of tab travel. Example: If a tab travels 15 degrees from the neutral or zero position to one extreme deflection, the transmitter shall be connected so that the transmitter shaft rotates 30 degrees for the 15 degrees tab travel.

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6.1.2 The trim tab indicator may be used with the AN5787 position transmitter which has the same electrical characteristics as in the AN5786 position transmitter. The transmitter brushes must be rotated 4 mechanical degrees for each 1 degree of tab travel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity desired.
(c) Selection of applicable levels of preservation and packaging, and packing. (See Section 5).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 and information pertaining to qualification of products may be obtained from that activity.

* 6.4 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Reclaimed materials. The use of reclaimed materials should be encouraged to the maximum extent possible.

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MIL-I-7063B Indicator, Position, Rudder Trim Tab

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